

Lab 1: Requirements Engineering and UML Use Case Modelling

Problem Statement 3: Campus Placement and Internship Pipeline

Student Name: Tanisha Mathur

SRN: PES1UG24CS491

Section: 5 I

System Actors and Identified Use Cases

Actors

- 1 Student Applicant, the primary actor who applies to drives and views application status.
- 2 Placement Officer, the primary actor who sets eligibility rules, manages interviews, and issues offers.
- 3 Document Parser Service, a secondary system actor that extracts text from uploaded resumes.

Use Cases

- UC 01: Apply for Placement Drive
- UC 02: Parse Resume and Extract Skills
- UC 03: Validate Eligibility Criteria
- UC 04: Claim Special Category Relaxation
- UC 05: Manage Interview Queues
- UC 06: Issue Offer Letters
- UC 07: Configure Drive Rules

Key Relationships

- UC 01, Apply for Placement Drive, includes UC 03, Validate Eligibility Criteria.
- UC 03, Validate Eligibility Criteria, includes UC 02, Parse Resume and Extract Skills.
- UC 04, Claim Special Category Relaxation, extends UC 03, Validate Eligibility Criteria.

Use Case Flow Specification

Use Case ID	UC 01	Use Case Name	Apply for Placement Drive
Primary Actor	Student Applicant	Secondary Actors	Placement Officer, Document Parser Service

Preconditions

- 1 The student is registered and logged into the placement portal.
- 2 An active placement drive is currently open for applications.
- 3 The student has uploaded a resume in PDF format.

Postconditions

- 1 The application status changes to "Submitted" in the portal.
- 2 The student receives an application confirmation with a reference ID.

Main Success Scenario

- 1 Student logs in and navigates to the active drives page.
- 2 System displays the list of open placement drives with their eligibility criteria.
- 3 Student selects a drive and clicks on Apply.
- 4 System invokes the eligibility check, which includes Validate Eligibility Criteria.
- 5 System uses the parser service to read the resume, which includes Parse Resume and Extract Skills, and compares the student's CGPA, branch, and backlog details against the drive rules.
- 6 System verifies that all criteria are satisfied.
- 7 System shows an application summary and asks for final confirmation.
- 8 Student clicks Confirm and Submit.
- 9 System saves the application, assigns an application ID, and updates the status.
- 10 System displays a success message on the screen.

Alternate Flows

4a. Ineligible Student

- 1 System detects that the student does not meet the minimum CGPA or has active backlogs.
- 2 System displays an error message stating the exact reason for ineligibility.
- 3 System prevents submission and returns the student to the dashboard.
- 4 Use case ends.

5a. Unreadable Resume PDF

- 1 System cannot extract text from the uploaded PDF resume.
- 2 System prompts the student to re upload a clean, machine readable PDF resume.
- 3 Student uploads a valid file.
- 4 System resumes the flow from Step 5.